SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM University, Greater Noida, U.P.

URL SHORTENER APPLICATION

Python Developer Internship Report

Submitted in partial fulfilment of the requirement for the degree of

B.Tech (Computer Science and Engineering)

Name of Student:	Prince Maurya
Roll Number:	25SCS1003001538
Programme:	B.Tech CSE
Semester / Section:	3rd Semester / Section 2CSE7
Batch:	2025-2029
Industry Partners:	upSkill Campus & UniConverge Technologies Pvt. Ltd. (UCT)
Collaboration:	AICTE

Certificate ID: USC838112TIA

Internship Period: 12-08-2026 to 02-09-2026

Academic Session: 2025-2029

CANDIDATE'S DECLARATION

I, **Prince Maurya**, hereby declare that this internship report titled **"Python Developer Internship - URL Shortener Application"** has been prepared by me as part of my academic internship requirement for the B.Tech Computer Science and Engineering programme at IILM University, Greater Noida.

This report presents my original internship experience, learning activities, technical implementation, testing, and project work carried out during the Python developer internship with upSkill Campus and UniConverge Technologies Pvt. Ltd. (UCT) in collaboration with AICTE.

I understand that I am responsible for the accuracy and authenticity of the material submitted in this report.

Student Name	Prince Maurya
Roll / Enrollment No.	25SCS1003001538
Programme	B.Tech CSE
Semester & Section	3rd Semester, Section 2CSE7
Batch	2025-2029
Date	September 2, 2026

Signature of Student
Prince Maurya

ACKNOWLEDGEMENT

I express my sincere gratitude to **IILM University, Greater Noida**, and the **School of Computer Science and Engineering** for providing the academic framework and opportunity to complete my industrial internship as part of the B.Tech programme.

I am deeply thankful to **upSkill Campus** and its industry partner **UniConverge Technologies Pvt. Ltd. (UCT)**, along with the **All India Council for Technical Education (AICTE)**, for providing this Python internship opportunity and practical exposure through hands-on training and industry-oriented project work.

I extend my gratitude to **Kaushlendra Singh Sisodia** (Chief Mentor at The IoT Academy & Director at UniConverge Technologies) and the mentorship team for guiding us through structured weekly milestones. Finally, I thank my faculty coordinators, family, and peers who supported me throughout this learning journey.

Prince Maurya
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Section: 2CSE7 | B.Tech CSE

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1. INTERNSHIP OVERVIEW

This report documents the industrial Python developer internship completed by **Prince Maurya**. The internship program was offered by **upSkill Campus** in collaboration with industry partner **UniConverge Technologies Pvt. Ltd. (UCT)** and supported by the **All India Council for Technical Education (AICTE)**.

As per official records, the internship commenced on **12-08-2026** and was successfully completed on **02-09-2026** under Certificate ID **USC838112TIA**. The primary domain of training and project delivery was Python Programming.

ACADEMIC INSTITUTION	IILM University, Greater Noida
DEGREE & BRANCH	B.Tech Computer Science and Engineering (CSE)
STUDENT DETAILS	Prince Maurya | Roll No: 25SCS1003001538 | Sec: 2CSE7
INTERNSHIP ROLE	Python Developer Intern
INDUSTRY PARTNERS	upSkill Campus & UniConverge Technologies Pvt. Ltd. (UCT)
GOVERNMENT PARTNER	AICTE (All India Council for Technical Education)
CERTIFICATE ID	USC838112TIA
DURATION	12-08-2026 to 02-09-2026
PROJECT TITLE	URL Shortener Application (GUI + Data Persistence)

2. ORGANIZATION & PROGRAM PROFILE

2.1 UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies (UCT) is an industry leader in Digital Transformation, specializing in Industrial IoT (IIoT), Cloud Computing, Embedded Systems, and Machine Learning solutions. UCT builds robust products like UCT Insight and Factory Watch, driving industrial automation and sustainable smart factory ecosystems.

2.2 upSkill Campus & The IoT Academy

upSkill Campus is an advanced career development and upskilling platform providing executive mentoring, self-paced technical modules, and real industry project exposure. In partnership with AICTE, upSkill Campus bridges the gap between college curricula and core industrial requirements.

3. INTERNSHIP OBJECTIVES

> The internship objectives were aligned with developing practical Python programming ability and applying those concepts to a working, deployable project.

- **Core Fundamentals:** Strengthen Python programming logic, object-oriented concepts, data structures, and modular code architecture.
- **GUI Development:** Understand Event-Driven Programming by transitioning from sequential terminal scripts to fully graphical applications using the `tkinter` library.
- **Data Persistence:** Master local file I/O operations and serialization using the `json` module to achieve permanent data storage across application reboots.
- **Defensive Programming:** Implement robust input validation, exception handling (`try-except` blocks), and error prevention mechanisms to ensure application stability.
- **System Integration:** Build a complete, user-friendly Desktop URL Shortener Application featuring direct system clipboard integration.
- **Version Control:** Manage code repositories using Git and publish professional project artifacts on GitHub.

4. LEARNING PROGRESS & WEEKLY MILESTONES

TIMELINE	KEY MILESTONES & TECHNICAL ACHIEVEMENTS
Week 01	**Python Core & Logic:** Project track selection (URL Shortener). Built foundational CLI logic in Python utilizing `string` and

TIMELINE	KEY MILESTONES & TECHNICAL ACHIEVEMENTS
	`random` libraries to generate secure 6-character short codes mapped via dictionaries.
Week 02	**Data Persistence:** Upgraded from volatile in-memory storage to persistent JSON storage using `json.dump()` and `json.load()`. Added strict URL validation (`http://`, `https://`) and an interactive continuous CLI loop.
Week 03	**GUI Implementation:** Transitioned from CLI to Graphical User Interface (GUI) using `tkinter`. Created window layouts, input boxes, action buttons, and integrated system clipboard functionality (`root.clipboard_append`).
Week 04	**Testing & Finalization:** UI refinement, edge-case validation, exception handling with `messagebox` alerts, implemented "Clear All" functionality, conducted end-to-end testing, and finalized GitHub documentation.

5. PROJECT DESCRIPTION - URL SHORTENER

The **URL Shortener Application** is a standalone Python desktop software designed to compress long, complex web addresses into clean, manageable 6-character alphanumeric links. In academic and professional environments, long URLs are cumbersome to share in documents, presentations, or emails, often leading to formatting breaks.

While commercial services like Bitly exist, they operate as cloud SaaS platforms requiring paid subscriptions for advanced features, internet connectivity for database retrieval, and they inherently track user data. This project solves that problem by offering a **100% offline, private, and lightweight desktop alternative**.

The system stores all URL mappings locally in a JSON database (`url_database.json`), ensuring that generated short links persist indefinitely without any reliance on external cloud infrastructure.

6. PROJECT FEATURES & MODULES

- **URL Shortening Engine:** Instantly accepts long URLs, verifies prefix validity, and generates unique 6-character short links.
- **Collision Prevention Algorithm:** Automatically executes a conditional `while` loop to cross-reference newly generated short codes against existing database keys, guaranteeing zero duplicates.
- **Original Link Retrieval:** Allows users to input a previously generated short code or shortened URL to instantly fetch the original destination link.
- **System Clipboard Copy:** Features a dedicated UI button that copies the generated short link directly to the operating system's clipboard for immediate pasting.
- **Persistent Local Storage:** Seamlessly reads from and writes to a local JSON file on application startup and state changes.

- **Clear Interface Action:** A utility function that resets all GUI entry fields and status messages with a single click.

7. TECHNOLOGY STACK

TECHNOLOGY / LIBRARY	PURPOSE & ROLE IN PROJECT
Python 3.x	The core programming language handling all backend application logic and control flow.
Tkinter	The standard Python GUI toolkit utilized for designing the window layout, interactive buttons, and text entry fields.
JSON Module	Handles data serialization to encode Python dictionaries into a `.json` file for permanent disk storage.
OS Module	Manages vital file system checks (`os.path.exists`) to prevent the application from crashing if the database file is missing.
String & Random	Responsible for the cryptographic generation of the 6-character alphanumeric combinations using `ascii_letters` and `digits`.
Git & GitHub	Employed for version control, code tracking, and professional portfolio hosting.

8. SYSTEM ARCHITECTURE & WORKING

The application follows a modular, multi-tier architecture to ensure clean separation of concerns:

1. **User Interface Tier (Tkinter GUI):** Captures user input (URLs or Short Codes) via text fields and triggers specific Python functions when action buttons are clicked.
2. **Validation Tier:** Before processing, the input is checked for empty strings and validated to ensure the presence of proper `http://` or `https://` protocol headers.
3. **Algorithmic Execution Tier:** If valid, the system generates random 6-character strings and executes the collision-check loops to ensure uniqueness.
4. **Storage Tier (JSON Database):** The new mapping is appended to the active Python dictionary and immediately serialized (saved) to `url_database.json`.
5. **Feedback Tier:** The application updates the GUI labels with the result, triggers pop-up alerts for success or errors, and prepares the OS clipboard for copying.

9. DATABASE DESIGN

Data persistence is managed via a flat-file JSON structure. The database operates as a simple key-value store, where the unique 6-character short code acts as the primary key mapped to the original target URL string.

```
{
    "aB9xK1": "https://www.iilm.edu/academics/computer-science",
    "7mP2qL": "https://github.com/Princemau25/url-shortener",
    "xY3z8N": "https://aicte-india.org"
}
```

10. IMPLEMENTATION DETAILS

The complete backend logic is encapsulated in a highly organized Python script. The core functional modules include:

- `load_database()` & `save_database()`: Safe JSON read/write handlers explicitly wrapped in `try-except` blocks to prevent application failure due to OS permission issues or corrupted files.
- `is_valid_url(url)`: A strict string validator ensuring that all inputs are properly formatted as web links.
- `shorten_url()`: The main operational function that generates keys, checks for collisions, executes the save command, and updates the GUI variables.
- `retrieve_url()`: Parses user input (handling both full short links and raw codes) to query the database dictionary and retrieve the original URL.
- `copy_clipboard()`: Invokes `root.clipboard_clear()` and `root.clipboard_append()`, followed by `root.update()` to force the OS to register the new clipboard state.

13. TESTING & RESULTS

Comprehensive end-to-end testing was conducted to ensure application stability, particularly focusing on edge cases and malformed user inputs.

TEST CASE SCENARIO	INPUT ACTION	EXPECTED & OBSERVED RESULT	STATUS
Empty Input Handling	Click 'Shorten' with a blank input field	Application intercepts the action; a `messagebox` warns: "Please enter a URL to shorten." No crash occurs.	**PASS**
Invalid URL Prefix	Enter "google.com" without http/ https	Intercepted by `is_valid_url()`. Alert thrown reminding user of protocol requirements.	**PASS**
Valid Link Generation	Enter valid long URL and submit	Generates 6-character short link, updates GUI, and successfully writes to the JSON file.	**PASS**
Clipboard Integration	Click 'Copy to Clipboard' button	Short link is copied to OS memory; success notification shown. System paste (Ctrl+V) works perfectly.	**PASS**
Database Survival	Restart application completely	Previously generated URLs are successfully loaded from JSON and retrieved without error.	**PASS**

14. GITHUB REPOSITORY & SUBMISSION

The complete project source code, JSON database sample, and detailed `README.md` documentation have been published to GitHub as required by the internship guidelines.

> **GitHub Repository Link:**
> https://github.com/Princemau25/url-shortener

Official Repository Naming Convention:

`Session_Name_RollNo_Semester_Section`

(e.g., `2025-2029_Prince_Maurya_25SCS1003001538_3rd_sem_2CSE7`)

15. SKILLS & LEARNING OUTCOMES

This internship provided substantial technical growth and practical exposure to professional software development practices:

- **Applied GUI Development:** Mastered the `tkinter` library to build event-driven window layouts, moving beyond basic sequential terminal logic.
- **Data Management:** Gained hands-on experience with data persistence techniques, utilizing JSON serialization and OS file path handling.
- **Software Development Life Cycle (SDLC):** Experienced iterative development through 4 weekly structured sprints, progressing from concept to a polished application.
- **Defensive Programming:** Developed strong habits for implementing rigorous input validation and graceful exception handling.
- **Professional Tooling:** Enhanced technical documentation, version control (Git), and repository management skills on GitHub.

16. CHALLENGES & FUTURE SCOPE

Challenges Overcome:

The primary challenge during development was managing the GUI layout using Tkinter's `pack` and `grid` managers to ensure elements did not overlap. Additionally, clipboard state synchronization posed an issue where the OS retained old data. This was resolved by forcing a `root.clipboard_clear()` and `root.update()` command cycle prior to appending new data.

Future Work Scope:

- **Database Migration:** Upgrading from local flat-file JSON storage to a robust cloud-based NoSQL database like Firebase or MongoDB for cross-device synchronization.

- **Web Deployment:** Refactoring the backend logic into a Flask or FastAPI framework to deploy the tool as a fully accessible web application.
- **Analytics Dashboard:** Implementing a click-tracking feature to monitor link usage, geographical access metrics, and link expiration parameters.

17. CONCLUSION & REFERENCES

The Python internship provided invaluable practical exposure, effectively bridging academic classroom concepts at IILM University with industry software practices driven by upSkill Campus and UCT. Designing and deploying the URL Shortener Application consolidated programming logic, GUI development, file persistence, and testing methodologies into a highly cohesive and professional portfolio project.

References

1. upSkill Campus & UniConverge Technologies (UCT) Internship Program Guidelines (2026).
2. AICTE Virtual Internship Domain Documentation - Python Programming.
3. Python Standard Library Documentation: Tkinter, JSON, Random, and String modules.
4. Official Internship Completion Certificate - Certificate ID: USC838112TIA.

ANNEXURE I

Internship Completion Certificate

CERTIFICATE OF COMPLETION

Issued by upSkill Campus, UniConverge Technologies (UCT) & AICTE

Certificate ID:	USC838112TIA
Awarded To:	**Prince Maurya**
Internship Domain:	Python
Internship Duration:	12-08-2026 to 02-09-2026

This program was conducted in collaboration with the **All India Council for Technical Education (AICTE)**, **UniConverge Technologies Pvt. Ltd. (UCT)**, and **upSkill Campus**. During this internship, the candidate actively

participated in hands-on training and real industry project work in the domain of Python.

(The original certificate has been verified and attached as part of this academic submission)